COLUMBIA UNIVERSITY

DEPARTMENT OF BIOSTATISTICS

P8100 – APPLIED REGRESSION I

Exercise Sheet 4 (Model Answers)

Question 1 [2+2+2+2+2+2 = 12 marks]

(a) The fitted model with the full data and all 6 predictors has ANOVA table:

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	14164.953	6	2360.826	10.195	<.001[b]
	Residual	7872.949	34	231.557		
	Total	22037.902	40			

a. Dependent Variable: Y

b. Predictors: (Constant), X6, X3, X4, X1, X5, X2

and regression coefficients

Coefficients[a]

Model		Unstandardized Coefficients				Collinearity Statistics	
		B	Std. Error	t	Sig.	Tolerance	VIF
1	(Constant)	60.921	38.694	1.574	.125		
	X1	-.874	.601	-1.453	.155	.306	3.265
	X2	.066	.016	4.033	<.001	.068	14.711
	X3	-.042	.016	-2.685	.011	.071	14.173
	X4	-.087	.159	-.544	.590	.962	1.039
	X5	.386	.369	1.046	.303	.307	3.255
	X6	-.002	.166	-.013	.990	.299	3.346

a. Dependent Variable: Y

(b) Based on the VIF's in the table above, the is multicollinearity between X_2 and X_3 (VIF >10). Since X_2 has the largest VIF, it is removed.

(c) X_2 is removed and the regression re-run. The ANOVA table is:

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	10398.623	5	2079.725	6.254	<.001[b]
	Residual	11639.279	35	332.551		
	Total	22037.902	40			

a. Dependent Variable: Y

b. Predictors: (Constant), X6, X3, X4, X1, X5

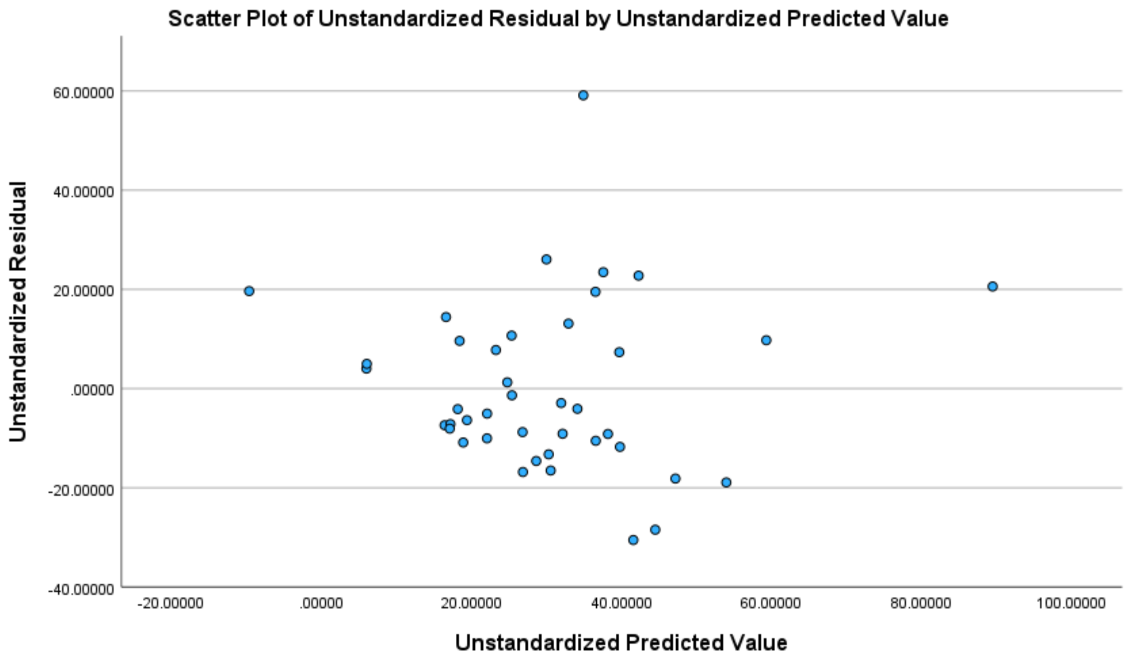
The regression coefficients are:

Coefficients[a]

Model		Unstandardized Coefficients		t	Sig.	Collinearity Statistics	
		B	Std. Error			Tolerance	VIF
1	(Constant)	89.536	45.584	1.964	.057		
	X1	-1.614	.686	-2.351	.024	.338	2.961
	X3	.019	.005	3.758	<.001	.991	1.009
	X4	-.129	.190	-.680	.501	.967	1.035
	X5	.533	.440	1.212	.234	.310	3.223
	X6	.010	.199	.051	.960	.299	3.345

a. Dependent Variable: Y

The residual plots is:



The topmost point seems to be an outlier.

(d) The studentized residuals are:

City	Studentized_Residuals
Phoenix	1.50149
Little Rock	-.36818
San Francisco	-.57986
Denver	-.29331
Hartford	1.14671
Wilmington	.60090
Washington	-.16044
Jacksonville	-.24031
Miami	.26687
Atlanta	-.07586
Chicago	1.84319
Indianapolis	-.65252
Des Moines	-.76247
Wichita	-.61708
Louisville	-.22524
New Orleans	-.43281
Baltimore	.40944
Detroit	-1.10253
Minneapolis St Paul	-1.05580
Kansas City	-.93714
St Louis	1.47513
Omaha	-.82692
Albuquerque	.30566
Albany	.74605
Buffalo	-1.91397
Cincinnati	-.50624
Cleveland	1.35957
Columbus	-.58554
Philadelphia	.58133
Pittsburgh	1.36462
Providence	4.13448
Memphis	-.74318
Nashville	-.48805
Dallas	-.46161
Houston	-.99951
Salt Lake City	.55779
Norfolk	.43137
Riebmond	.06957
Seattle	-.54940
Charleston	1.02323
Milwaukee	-1.66395

The cut-off from t-tables for d.f. = 41 – (5+2) = 34 is 2.03. Based on this cut-off, Providence is an outlier and is removed.

(e) With Providence removed, the fitted model is:

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	10100.640	5	2020.128	8.868	<.001[b]
	Residual	7745.260	34	227.802		
	Total	17845.900	39			

a. Dependent Variable: Y

b. Predictors: (Constant), X6, X3, X4, X1, X5

Coefficients[a]

Model		Unstandardized Coefficients B	Std. Error	t	Sig.	Collinearity Statistics Tolerance	VIF
1	(Constant)	53.463	38.724	1.381	.176		
	X1	-1.034	.585	-1.768	.086	.324	3.090
	X3	.021	.004	5.034	<.001	.989	1.011
	X4	-.100	.158	-.636	.529	.965	1.036
	X5	.195	.373	.522	.605	.297	3.364
	X6	.123	.167	.737	.466	.292	3.421

a. Dependent Variable: Y

(f) We fit a stepwise regression model with $\alpha_{\text{IN}} = .05$ and $\alpha_{\text{OUT}} = .051$[*]. The final model is

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
2		9181.178	2	4590.589	19.603	<.001[c]
		8664.722	37	234.182		
		17845.900	39			

a. Dependent Variable: Y

c. Predictors: (Constant), X3, X1

[*] $\alpha_{\text{OUT}} = .051$ is not a rule. It is also possible to use $\alpha_{\text{OUT}} = .1$

Coefficients[a]

Model: 2

	Unstandardized Coefficients		t	Sig.
	B	Std. Error		
(Constant)	77.430	19.464	3.978	<.001
X3	.021	.004	5.054	<.001
X1	-1.112	.339	-3.285	.002

a. Dependent Variable: Y

Question 2 [8 marks]

The following variables are defined.

- n = total number
- x = number taking drugs
- ill = 1 if subject is ill, =0 otherwise
- gender = 1 if female, =0 otherwise
- z_1= 1 if age group is 35-54, =0 otherwise
- z_2 = 1 if age group is ≥ 55, = 0 otherwise (age group ≤ 34 is therefore the reference group)†

The data look as follows:

† Other definitions are also possible, e.g. z_1 =1 if in ≤ 34 age group, = 0 otherwise; z_2 =1 if in 35-54 age group, = 0 otherwise – this makes the ≥ 55 age group become the reference group. Similarly, for ill and gender, the dummy variables can be reversed.

n	x	ill	gender	z1	z2
27	1	1	0	0	0
47	1	1	0	1	0
127	9	1	0	0	1
42	4	1	1	0	0
116	26	1	1	1	0
164	32	1	1	0	1
152	1	0	0	0	0
138	2	0	0	1	0
90	1	0	0	0	1
140	1	0	1	0	0
109	1	0	1	1	0
71	5	0	1	0	1

If p is the probability of drug use, then the following logistic regression is run:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 \text{ill} + \beta_2 \text{gender} + \beta_3 z_1 + \beta_4 z_2$$

The table of model effects is:

Tests of Model Effects

Source	Type III		
	Wald Chi-Square	df	Sig.
(Intercept)	234.308	1	<.001
ill	32.050	1	<.001
gender	19.003	1	<.001
z1	3.996	1	.046
z2	5.416	1	.020

Events: x
Trials: n
Model: (Intercept), ill, gender, z1, z2

The table of parameter estimates is:

Parameter Estimates

Parameter	B	Std. Error	Hypothesis Test		Exp(B)	95% Wald Confidence Interval for Exp(B)	
			df	Sig.		Lower	Upper
(Intercept)	-5.560	.4969	1	<.001	.004	.001	.010
[ill=1]	1.938	.3423	1	<.001	6.945	3.550	13.584
[ill=0]	0[a]	.	.	.	1	.	.
[gender=1]	1.303	.2988	1	<.001	3.679	2.048	6.607
[gender=0]	0[a]	.	.	.	1	.	.
[z1=1]	.882	.4414	1	.046	2.417	1.017	5.740
[z1=0]	0[a]	.	.	.	1	.	.
[z2=1]	1.006	.4324	1	.020	2.736	1.172	6.385
[z2=0]	0[a]	.	.	.	1	.	.
(Scale)	1[b]						

Events: x
Trials: n
Model: (Intercept), ill, gender, z1, z2

a. Set to zero because this parameter is redundant.

b. Fixed at the displayed value.

Based on these results, adjusting for age group and gender, subjects who are ill have a significantly greater odds of drug use by a factor 6.945 [OR = 6.945, 95% CI = (3.550, 13.584), pval < .001]